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efficiency. LEDs are typically coupled to a heat sink to minimise the junction temperature. Limiting the current or power through an AC LED helps prevent thermal runaway and provides the added benefit of better brightness regulation over a wide temperature range, as well as longer lifetime.

The AC LED consists of a single LED and a combination of resistors that are used to set the operating current based on its forward voltage. The LED consists of two sets of diodes of opposite polarity connected in parallel as shown in Fig. 1. As the line voltage increases on the positive half cycle, one of the internal strings conducts to generate light. During the negative half cycle, the other string conducts to generate light.

Fig. 2 shows the AC LED test set-up, where an MOV and a surface-mount PPTC device (rated at 60V with a 140mA hold current) are integrated on the AC LED lighting system board. The table shows the test parameters and results.

Over-voltage protection design considerations

AC LEDs are being used increasingly in general-purpose illumination lighting systems for indoor and outdoor use. With AC mains as their direct power supply, lightning may cause transient over-voltage conditions on the AC line. Without robust and reliable circuit protection, surge or ring waves can damage the AC LED. Because changes in voltage can produce a disproportionate change in current and affect the light output, the LEDs must be protected from these AC line-voltage fluctuations.

The PPTC device helps limit the current flowing through the LED during steady-state over-current operation, protecting it from voltage fluctuations. MOVs are used for transient over-voltage suppression in AC line voltage applications.

Figs 3 and 4 show the test results on the AC LED with a 7mm, 200V MOV. As seen from the results, the MOV solves lightning and ring wave issues. Also, the light output is very stable during lightning surge and ring wave testing. This approach helps equipment remain operational after specified lightning tests according to IEC 61000-4-5 and ring wave tests according to IEEEC.62.41.

To sum up

Pairing a PPTC over-current/over-temperature protection device with an MOV provides a completely resettable and coordinate circuit protection solution for AC LED lighting systems.

New products in development by manufacturers include over-current, over-voltage and over-temperature protection provided by a PPTC/MOV combination device. The hybrid device combines a PPTC element with an MOV in a single thermally-protected device, giving designers the flexibility of integrated protection in a single device. ●

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